

ADDING INTEGERS USING INTEGER CHIPS



LESSON INTRODUCTION

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

LEARNING TARGETS

- I can express integer addition using integer chips.
- I can add integers using integer chips.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.2 Add, subtract, multiply, and divide multidigit numbers.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What is an integer?
- How can you use integer chips to represent integer addition?
- What is a zero pair?
- How do you use integer chips to add integers?

LESSON NARRATIVE

In this lesson, students represent integer addition using integer chips to begin developing an understanding of how to add signed numbers. Students explore zero pairs and begin generalizing about patterns they observe when adding integers with the same sign or opposite signs.

COMPONENT SUMMARY

7.3.1 WARM-UP (~5 MIN.)

Make connections between elevations and positive and negative integers

7.3.3 EXPLORATION (10 MIN.)

Explore adding integers with the same sign using integer chips

7.3.5 EXPLORATION (15 MIN.)

Explore using zero pairs to add integers with opposite signs using integer chips

7.3.7 WRAP-UP (~5 MIN.)

Demonstrate understanding of the lesson learning targets

7.3.2 GUIDED INSTRUCTION (5-10 MIN.)

Represent simple money contexts with positive and negative integers and integer chips and define zero pairs

7.3.4 BRING IT TOGETHER (5 MIN.)

Summarize how to add integers with the same sign using integer chips

7.3.6 BRING IT TOGETHER (5 MIN.)

Summarize how to add integers with opposite signs

RESOURCES

GLOSSARY

Key Terms:

- Integers
- Zero pair
- Zero pair as a mathematical equation

Additional Terms: N/A

MATERIALS

- Integer chips
- (Optional) \$1 manipulatives

ADDITIONAL PREPARATION

- Consider creating sentence stems and/or answer choices for question 2 in 7.3.6 Bring It Together.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to come up with real-world scenarios to represent addition expressions, particularly when addends are negative.

THINGS TO CONSIDER

- Consider providing integer chips for students in 7.3.5 Exploration or encouraging students to draw a representation if they get stuck.
- Consider using \$1 manipulatives in 7.3.3 Exploration to help students connect the idea of spending and earning money to positive and negative integers.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 7.3.3 Exploration and 7.3.5 Exploration, provide an opportunity for students to notice and wonder about patterns in the sums when adding integers with the same or opposite signs.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 7.3.3 Exploration and 7.3.5 Exploration, students have opportunities to represent adding integers using multiple representations. Students make connections among written descriptions of real-world scenarios, numeric expressions, and integer chips.

DIFFERENTIATE INSTRUCTION

Lesson 3 incorporates integer chips into instruction to provide meaningful learning strategies for making sense of adding integers. Provide integer chips for students to explore each activity throughout the lesson.

Earn	Spend
	

7.3.1 WARM-UP: BELOW SEA LEVEL

Component Overview: Students connect positive and negative integers to distances above and below sea level and sketch an example.

QUESTIONING

Dr. Ernest Everett Just accomplished many things in his career that may pique students' interests in the STEM field. Consider an interdisciplinary extension project with a science teacher on one of his many roles.

ENRICHMENT

As students work on question 1, consider asking them the following:

- Does your sketch more closely resemble a horizontal or vertical number line? Why?
- What value does sea level represent in your sketch?



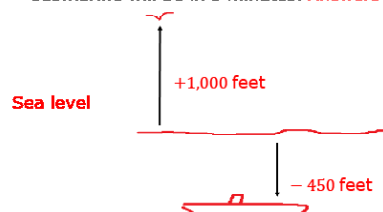
7.3.1 Warm-Up: Below Sea Level

Dr. Ernest Everett Just was a pioneering African American biologist who conducted research at Woods Hole Marine Biological Laboratory in Massachusetts. He published two books and more than 70 scientific papers, most of which focused on the cell. Today, marine biologists use submarines to study marine life.



A submarine explorer begins at sea level and descends at a rate of 150 feet per minute.

- Sketch a picture that shows how far below sea level the submarine will be in 3 minutes. *Answers will vary.*



- Add a bird that is flying 1,000 feet above sea level to your picture. *See above*
- What number represents sea level?
0
- The location of the submarine is represented by a
☐ positive value,
☒ negative value,
 represented by a
 while the location of the bird is
☐ positive value.
☒ negative value.



7.3.2 Guided Instruction: Zero Pairs

Positive and negative values are often useful in the contexts of money, banking, and accounting.

- In the context of money, o positive
o negative numbers can be used to represent money spent or money owed to someone.
- o Positive
o Negative numbers can be used to represent money earned.

Money can be used to learn how to add integers.

- A value of $+1$ means that you **earned** \$1.
- A value of (-1) means that you **spent** \$1 or **owe** \$1.

The table below shows a positive integer chip, which represents earning \$1, and a negative integer chip, which represents spending \$1.

Earn	Spend

Integer chips can also be used to represent integers.



Integers are whole numbers and their opposites.

A positive number can be written with a positive sign, $+1$ or $(+1)$, or it can be written without any sign, 1.

A negative number can be written as -1 , (-1) or $\bar{1}$.

- Complete the table. In the last column, sketch the integer chips that represent the phrase and the integer. Place a $+$ sign in a circle to represent and place a $-$ sign in a circle to represent .

Phrase	Integer	Integer Chips
Earned \$6	$(+6)$	
Spent \$3	-3	

- Complete the table by sketching integer chips for each phrase.

Earned \$4	Spent \$4	Earned \$4, Then Spent \$4

- Joleen earned \$4, then spent \$4. How much money does she have left?
0

The last column in the table and problem above both represent a **zero pair**.

A **zero pair** is a pair of numbers whose sum is zero.

7.3.2 GUIDED INSTRUCTION: ZERO PAIRS

Component Overview: Students learn how positive and negative integers can be represented using integers and integer chips. The concept of zero pairs is introduced in the context of adding integers. Students review the definition and then use zero pairs to add integers.

DIFFERENTIATE INSTRUCTION

Consider providing a few real-world examples of positive and negative numbers representing money. These examples will activate prior knowledge and help students connect spending and earning to positive and negative numbers.

TEACHER MOVES

Consider the tools available for instruction during this activity. Modeling each example using inquiry would benefit students as they make connections from the integer-chip representations to their numerical representations.

ENRICHMENT

Consider asking students the following:

- Can you explain what a zero pair is in your own words?
- Can you provide an example of a zero pair?

7.3.2 GUIDED INSTRUCTION: ZERO PAIRS (CONTINUED)

TEACHER MOVES

Students are only given the visual model for positive integer chips. They complete the visual model for negative integer chips and the sum.

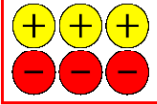
DIFFERENTIATE INSTRUCTION

To challenge advanced learners, consider having students write a word problem to go with the equations they have created to represent zero pairs.

TEACHER MOVES

Students are only given the visual model for negative integer chips. They complete the visual model for positive integer chips and the sum.

4. Complete the table by drawing the negative integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
		

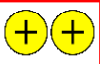
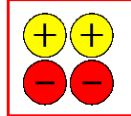
Each zero pair can be represented by an equation. The box shows two mathematical equations that represent the zero pair of  and .



Zero Pair as a Mathematical Equation

$$+1 + (-1) = 0 \quad \text{or} \quad 1 + -1 = 0$$

5. Write an equation that represents the **sum** column in the previous table.
 $+3 + (-3) = 0$
6. Complete the table by drawing the positive integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

Positive	Negative	Sum
		

7. Write an equation that represents the **sum** column in the previous table.
 $+2 + (-2) = 0$



7.3.3 Exploration: Adding Integers with the Same Sign

1. Work with your partner to complete the table.

Expression	Money Sentence	Integer Chips	Sum
$5 + 2$	I earn \$5 babysitting and another \$2 for cleaning the dishes.		7
$3 + 2$	Answers will vary. I earn \$3 mowing the lawn and another \$2 for taking out the trash. I earn \$3 washing dishes and another \$2 for babysitting.		5
$-4 + -3$	Answers will vary. I owe my mom \$4 and I spent \$3 at the dollar store. I loaned my friend \$4 and I spent \$3 on my mom's birthday present.		-7
$(-2) + (-1)$	Answers will vary. I spent \$2 at the candy store and I paid back my mom the \$1 I owed her. I bought a \$2 ice cream cone and I loaned my sister \$1.		-3

2. With your partner, discuss the patterns you notice about the signs of the sums when adding integers with the same sign.
Answers will vary. I notice that when the signs were the same, we added the total number of chips.
3. Were any zero pairs created when adding integers with the same sign? Explain.
Answers will vary. No zero pairs were created when adding integers with the same sign. Instead, the result was a larger quantity of the signed integers.

7.3.3 EXPLORATION: ADDING INTEGERS WITH THE SAME SIGN

Component Overview: Students work with a partner to complete a table that includes multiple representations of adding integers with the same sign. The table includes the numeric expression, money sentence, integer chips, and sum. Next, students draw conclusions about patterns they notice about the signs of the sums and if there are any zero pairs when adding integers with the same sign.

TEACHER MOVES

Students are only given the visual model of integer chips for the first row.

TEACHER MOVES

Notice and Wonder

Consider engaging students with a discussion regarding what they notice and wonder about the signs and if there were any zero pairs prior to them writing their responses to questions 2 and 3.

ENRICHMENT

Consider asking students the following:

- What is the sign of the sum of two positive integers?
- What is the sign of the sum of two negative integers?

7.3.4 BRING IT TOGETHER: ADDING INTEGERS WITH THE SAME SIGN

Component Overview: Students summarize how to add integers with the same sign using integer chips. Consider creating an anchor chart and displaying it for students to reference.

TEACHER MOVES

Prompt students to justify their visual models to a partner. Highlight any students that make changes based on their partner's feedback to encourage growth mindset.

ENRICHMENT

Consider asking students the following:

- Will the sum of two positive integers always be positive?
- Will the sum of two negative integers always be negative?
- How do you know?

QUESTIONING

To add complexity, consider asking students who finish early to write equations to match each visual model of addition.



7.3.4 Bring It Together: Adding Integers with the Same Sign

1. Complete the visual model of the rule of adding two positive integers by drawing the integer chip that represents the **sign** of the sum.

Visual Model of Addition	Sign of Sum
	

2. Complete the visual model of the rule of adding two negative integers by drawing the integer chip that represent the **sign** of the sum.

Visual Model of Addition	Sign of Sum
	



7.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs

It is possible to add a positive number and a negative number. To find the answer using integer chips, combine a and to create a zero pair. Make as many zero pairs as possible. Then, determine the number of chips that are left over.

- Complete the table to show how the sum $5 + (-4)$ can be represented using integer chips.

Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	4 zero pairs
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the sum using the number of integer chips remaining.	$5 + (-4) = 1$

- What do you notice about the remaining integer chips and the sign of the sum?
There was 1 positive integer chip remaining. The sum was 1.

7.3.5 EXPLORATION: USING ZERO PAIRS TO ADD INTEGERS WITH OPPOSITE SIGNS

Component Overview: Students use integer chips to represent the sum of integers with opposite signs, determining the sum based on the zero pairs. Students then work with a partner to continue adding integers with opposite signs and create money sentences to represent addition expressions. Students work independently to find the sum of integers, check their partner's work, and use the sums to generalize how to add integers with opposite signs.

ENRICHMENT

Consider asking students the following:

- How can you represent 5 with integer chips?
- How can you represent -4 using integer chips?
- Why is the sum of 5 and -4 positive?

TEACHER MOVES

For both questions 1 and 3, students are given the visual model for the first two steps and must complete the third row in addition to synthesis questions 2 and 4.

- Complete the table independently to show how the sum $4 + (-7)$ can be represented using integer chips. Check your work with your partner.

Step	Response
Represent the first addend with integer chips.	
Represent the second addend with integer chips.	
Determine the number of zero pairs created, if any.	4 zero pairs
Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model.	
Determine the sum using the number of integer chips remaining.	$4 + (-7) = -3$

- What do you notice about the remaining integer chips and the sign of the sum?
There were 3 negative integer chips remaining. The sum was -3.

7.3.5 EXPLORATION: USING ZERO PAIRS TO ADD INTEGERS WITH OPPOSITE SIGNS (CONTINUED)

TEACHER MOVES

Consider having students discuss what they think the sign of the sums will be based on what they notice about the addition expressions before having them complete the table for question 5. Guide students to make logical predictions based on what they know about adding signed numbers and absolute value.

DIFFERENTIATE INSTRUCTION

Consider having students who finish early represent each expression in an additional way, either with an equivalent expression, a visual representation, or a written scenario.

TEACHER MOVES

Turn and Talk

Ask students to verbalize their responses before completing their written explanation to questions 7 and 8. This will solidify student understanding and allow for any misconceptions to be resolved before the 7.3.7 Wrap-Up.

ENRICHMENT

What do you notice about the sum when the addend with the larger absolute value is negative?

5. Work with your partner to complete the table. The first row is completed.

Expression	Money Sentence	Integer Chips	Sign of Sum	Sum
$4 + (-2)$	I earn \$4 and spend \$2.		+	+2
$3 + (-6)$	I earn \$3 and spend \$6.		-	-3
$-7 + 5$	I spend \$7 and earn \$5.		-	-2
$(-1) + 8$	I spend \$1 and earn \$8.		+	+7

6. Complete the table below independently. Check your work with your partner.

Mathematical Expression	Number of Zero Pairs	Sign of Sum	Sum
$(-10) + 12$	10	+	2
$12 + -7$	7	+	5
$(-11) + 4$	4	-	-7
$8 + -8$	8	No sign	0
$-10 + 3$	3	-	-7
$9 + (-2)$	2	+	7

7. What do you notice about expressions where the sum is a positive number?

Answers will vary. The number with the larger absolute value in the mathematical expression is positive.

8. What do you notice about expressions where the sum is a negative number?

Answers will vary. The number with the larger absolute value in the mathematical expression is negative.



7.3.6 Bring It Together: Adding Integers with Opposite Signs

The table below shows visual models of rules for adding integers with opposite signs. In each visual model, the larger circle symbolizes the integer represented by the most integer chips.

- Write an example of the rule. The first row has been completed.

Visual Model of Addition	Sign of Sum	Example
		$2 + (-5) = -3$
		<i>Answers will vary.</i> $-3 + 10 = 7$ $-5 + 7 = 2$
		<i>Answers will vary.</i> $6 + (-4) = 2$ $10 + (-6) = 4$
		<i>Answers will vary.</i> $-9 + 8 = -1$ $-12 + 4 = 8$

- Give two observations about the rules for adding integers with opposite signs.
Answers will vary.
When adding integers with opposite signs:
 - Ignoring the signs of the numbers in the mathematical expression, subtract the numbers.
 - The sign of the sum will be the sign of the number with the greater absolute value in the mathematical expression.

7.3.6 BRING IT TOGETHER: ADDING INTEGERS WITH OPPOSITE SIGNS

Component Overview: Students write expressions to represent a visual model of addition of integers with opposite signs. Then, students explain two things they noticed about adding integers with opposite signs.

ENRICHMENT

Consider asking students the following:

- What do you notice about the size of these integer chips?
- How does the absolute value of each addend impact the sign of the sum?

TEACHER MOVES

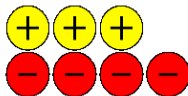
Common Misconception

Monitor for students who may articulate their answers with phrases like “the bigger number will always be larger” or “the answer will take the sign of the bigger number.” The sum will take the sign of the addend with the larger absolute value, which is not the larger number if it is negative.



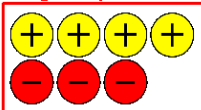
7.3.7 Wrap-Up: Ticket Out the Door

- Simone created the sketch below to represent the expression $4 + (-3)$.



Her sketch is incorrect. Explain why and create a new sketch to find the sum.

There should be 4 positive integer chips and 3 negative integer chips.



- Complete the table for the expression below.

Expression	$3 + -5$
Integer Chips	
Money Sentence	<i>Answers will vary. I earned \$3 mowing the lawn and spend \$5 on pizza.</i> <i>I earned \$3 taking out the garbage and spend \$5 on candy.</i>
Sign of the Sum	-
Sum	-2

7.3.7 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students demonstrate their understanding of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.

DIFFERENTIATE INSTRUCTION

Consider providing integer chips to provide a kinesthetic entry point.

Also, consider providing sentence starters for question 2 to help students who struggle writing explanations.